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Anthropometric analysis of 3D ear scans of Koreans and Caucasians for ear product design

Wonsup Lee^a , Xiaopeng Yang^b , Hayoung Jung^b , Ilgeun Bok^c, Chulwoo Kim^c ,
Ochae Kwon^c and Heecheon You^b 

^aSchool of Global Entrepreneurship and Information Communication Technology, Handong Global University, Pohang, South Korea;

^bDepartment of Industrial and Management Engineering, Pohang University of Science and Technology, Pohang, South Korea;

^cDesign Team, Mobile Communication Division, Samsung Electronics, Seoul, South Korea

ABSTRACT

The present study measured 25 dimensions of the ear including the concha and ear canal for ergonomic design of ear products and compared with existing ear measurement studies. Scanning and casting methods were employed to produce 3D ear images for 230 Koreans and 96 Caucasians and measurements of the ear dimensions were obtained by identifying 21 landmarks on individual ear scan image. The Korean ear measurements were found significantly larger (mean difference $\bar{d} = 0.4\text{--}3.7\text{ mm}$) and more varied (ratio of SDs = 1.01–1.55) than those of Caucasians in most of ear dimensions. The average ear length and ear breadth of male were significantly longer ($\bar{d} = 1.3\text{--}7.0\text{ mm}$) and wider ($\bar{d} = 0.8\text{--}3.0\text{ mm}$) than those of female. Use of gender- and ethnicity-composite ear data is recommended in product design due to the much larger intra-population variations (7.5–22.2 mm) than the corresponding inter-population variations.

Practitioner Summary: The 3D ear measurements of Koreans and Caucasians were collected and compared with those of different ethnic populations. The distinct ear features of the populations identified in this study are applicable to ergonomic design of ear products with better fit and comfort.

Abbreviations: CCW: cavum concha width; CV: coefficient of variation; EB: ear breadth; EL: ear length; SD: standard deviation; SE: sampling error; 3D: 3 dimensional

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Ear measurement; 3D anthropometric analysis; Korean ear; Caucasian ear; ear product design

1. Introduction

Understanding of the complex shape of the ear is needed for ergonomic design of wearable ear products. Most wearable ear products such as earphones, earmuffs, earplugs, and hearing aid devices interface with the outer ear, which comprises the pinna (auricle) and the ear canal (external acoustic meatus running from pinna to the middle ear). As shown in Figure 1, the concha located in the inner part of the pinna is separated into cymba concha and cavum concha by crus of the helix (Alvord and Farmer 1997), and the ear canal is curved with two bending points (Azernikov 2010; Pirzanski 2010; Sickel et al. 2011). The anatomical and anthropometric characteristics of the ear would be effectively utilised in designing ergonomic shapes of various ear products for fit and comfort (Jung and Jung 2003; Liu 2008).

A few studies measured ear dimensions of the concha in detail, but a comprehensive study on ear measurement including the ear canal inside has not been

conducted. Ahmed and Omer (2015), Bozkir et al. (2006), and Kalcioglu et al. (2003) measured the length and depth of the cavum concha using a conventional direct measurement for Sudanese adults (age: 18–30 years; male: 100 and female: 100), and Turkish adults (age: 18–25 years; male: 191 and female: 150) and children (age: 0–18 years; gender mixed: 1552) for both. Jung and Jung (2003) and Liu (2008) measured the ear, including ear canal length, for Korean adults (age: 17–89; male: 300 and female: 300) using a direct measurement method and for Taiwanese adults (age: 20–59; male: 100 and female: 100) using a photogrammetry measurement method, for application of ear-phone design. However, direct and photogrammetry measurement methods employed in ear measurement could reduce accuracy due to measurement errors caused by skin deformation and parallax distortion, which could be minimised in the 3D scanning measurement method (Lee et al. 2017). However, a comprehensive ear measurement study including more than

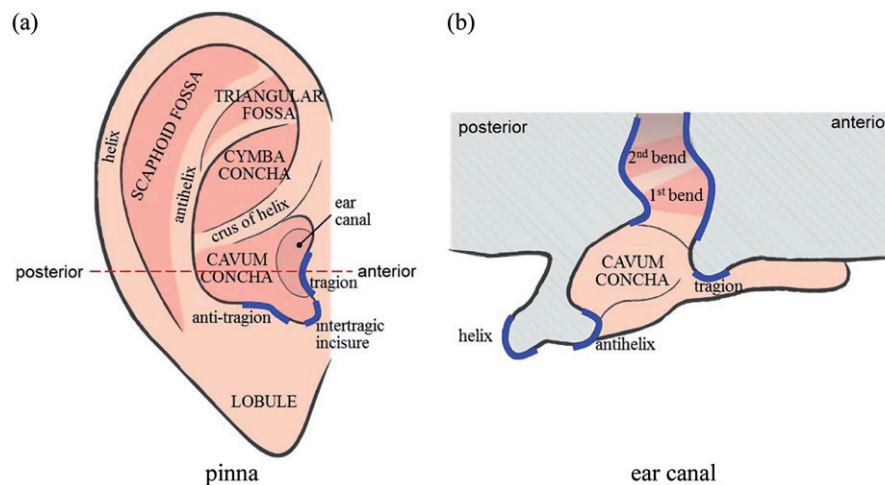


Figure 1. The anatomical structure of the ear.

20 ear dimensions has not been conducted. Therefore, an in-depth study, using 3D ear scans, is needed to measure ear dimensions such as concha depth, ear canal width, ear canal circumference, and ear canal angle with accuracy.

The 3D scanning technology is effectively applicable in the anthropometric analysis of complex shaped ear, once the ear is accurately scanned. The 3D scanning technology has been utilised to measure not only simple dimensions, such as length and circumference, but also complex dimensions such as shape, surface area, and volume for a body part under consideration (Chang et al. 2007; Lee et al. 2013a, 2017; Weinberg et al. 2004). Only partial areas of the ear, not including the ear canal and the pinna areas below the helix, anti-helix, and crus of helix, can be captured directly into 3D images by a 3D scanner (Jung et al. 2014; Lee et al. 2015). On the contrary, the sheltered areas of the pinna and the ear canal can be captured into 3D images by scanning their cast fabricated by a lightweight, malleable material, as employed by Azernikov (2010), Pirzanski (2010), and Sickel et al. (2011) to design a custom hearing aid for the ear canal.

The present study is intended to measure ears of Koreans and Caucasians using 3D ear scan images for ergonomic ear product design and compare their anthropometric characteristics with corresponding previous ear measurement studies. In total, 25 ear dimensions were selected for this study as those applicable to designing wearable ear products, and 3D ear scans were collected from 230 Koreans and 96 Caucasians for anthropometric analysis. Lastly, measurements of ear length, ear breadth, and cavum concha width of Koreans and Caucasians were compared with those selected ethnic groups such as Dutch, Nigerian, Sudanese, Taiwanese, and Turkish in terms of intra- and inter-population variabilities to examine the difference of ear size by ethnicity.

2. Methods

2.1. Selection of ear dimensions

In the present study, 25 ear dimensions and 21 landmarks as shown in Figures 2 and 3 were selected through literature review and discussion of a panel of experts in ergonomics and ear product design. In total, 19 ear dimensions (ear length, ear length above plane of trigion, ear connection length, ear breadth, tragus to helix width, ear protrusion, ear protrusion at superior auricle level, pinna rotation angle, pinna flare angle, lobular length, lobular width, lobule thickness, cavum concha length, cymba concha length, fossa length, cavum concha width, intertragal incisure width, cavum concha depth, and ear canal length) were found by reviewing 15 papers related to ear measurement (Ahmed and Omer 2015; Alexander and Laubach 1968; Algazi et al. 2001; Barut and Aktunc 2006; Bozkir et al. 2006; Buckley et al. 2005; Jung and Jung 2003; Jung et al. 2014; Kalcioglu et al. 2003; Kaushal and Kaushal 2011; Liu 2008; Meijerman, van der Lugt, and Maat 2007; Porter and Olson 2001; Taura, Adamu, and Modibbo 2013; Yoon and Jung 2002). Of the 19 ear dimensions, 9 dimensions were selected as shown in Figure 2 as those applicable to ear product design. Further, 14 dimensions were newly defined by a panel of two ergonomists in academia and three subject-matter experts in earset product design as those applicable to designing ear-related products such as open, in-ear, and ear-band type earphones, earplugs, and hearing aid devices by checking the relevance of each ear dimension to earset design dimensions. Lastly, 21 landmarks on the ear (Figure 3) were specified based on literature review and the panel discussion for measurement of 25 ear dimensions.

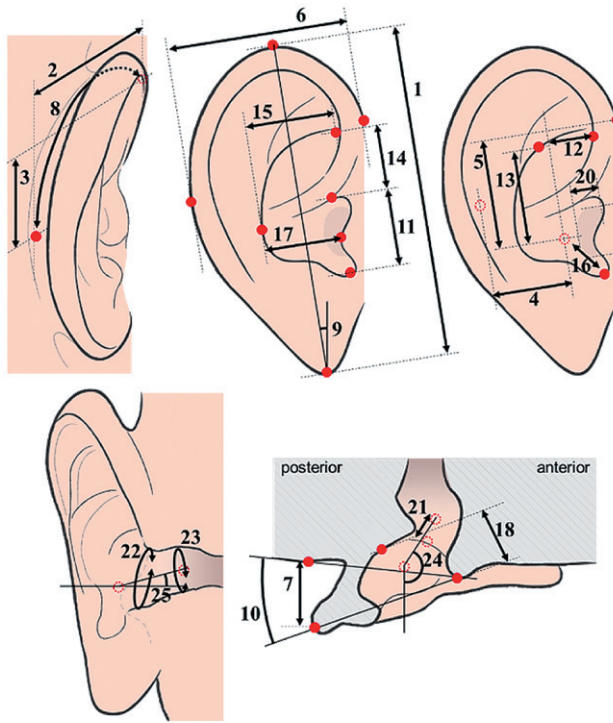


Figure 2. Ear dimensions related to wearable ear product design.

Category	No	Ear dimensions	LR	PD
Overall	1	ear length	○	
	2	otobasion superius to otobasion posterius horizontal length	○	
	3	otobasion superius to otobasion posterius vertical length	○	
	4	center of concha to otobasion posterius length	○	
	5	center of concha to otobasion superius length	○	
Width	6	ear breadth	○	
	7	ear protrusion	○	
Arc	8	upper otobasion arc	○	
Angle	9	ear angle	○	
	10	pinna flare angle	○	
Concha	11	cavum concha length	○	
	12	center of concha to anterior cymba concha length	○	
	13	center of concha to superior cymba concha length	○	
	14	superior cavum concha to anterior cymba concha length	○	
	15	posterior concha to anterior cymba concha length	○	
	16	center of concha to incisura intertragica length	○	
	17	cavum concha width	○	
Ear canal	18	cavum concha depth	○	
	19	ear canal entrance height	○	
	20	ear canal entrance width	○	
	21	ear canal entrance to 1 st bend length	○	
	22	ear canal entrance circumference	○	
	23	Ear canal 1 st bend circumference	○	
	24	ear canal azimuth angle	○	
	25	ear canal elevation angle	○	

2.2. Measurement of ear dimensions

2.2.1. Participants

Ear dimensions of 230 Koreans and 96 Caucasians were measured in the present study. The minimum sample size for Koreans was determined by Equation (1) (ISO 2012) using the Size Korea data (KATS 2004) with a 95% confidence level ($1-\alpha$). To calculate the coefficient of variation (CV) in Equation (1), the sample mean ($\bar{X}$) and sample standard deviation (s) of an ear dimension of the Korean population mixed in gender (male:female=1:1) and age groups (20s: 30s: 40s: 50s=1: 1: 1: 1) were estimated by Equations (2) and (3) (Lee et al. 2013a, 2013b), respectively, using the Size Korea data.

$$n = \left(1.96 \times \frac{CV}{k} \right)^2 \times 1.534^2 \quad (1)$$

$$\bar{X} = \frac{\sum_{i=1}^j \bar{X}_i \times n_i}{\sum_{i=1}^j n_i} \quad (2)$$

$$s = \sqrt{\frac{\sum_{i=1}^j [n_i \times \bar{X}_i^2 + (n_i - 1) \times s_i^2] - \sum_{i=1}^j n_i \times \bar{X}^2}{\sum_{i=1}^j n_i - 1}} \quad (3)$$

$$\cong \sqrt{\sum_{i=1}^j [p_i \times (\bar{X}_i^2 + s_i^2)] - \bar{X}^2}$$

where, CV=coefficient of variation, k =precision level, $\bar{X}$ =sample mean of a composite population,

s =sample SD of a composite population, $\bar{X}_i$ =sample mean of population i , n_i =sample size of population i , j =number of populations, s_i =sample SD of population i , p_i =proportion of population i .

Data of ear length and ear breadth provided in the Size Korea anthropometric survey were used to determine the minimum sample size for Koreans in the present study. Five levels of precision ($k=1-5\%$) were compared to determine a reasonable precision level regarding sampling error (SE) and minimum sample size as shown in Table 1. A precision level of 2% (maximum SE=1.2 mm in ear length and 0.7 mm in ear breadth) was recommended by the expert panel for ear product design as an acceptable error level, resulting in $n=106$ as the minimum sample size of the ear anthropometric survey. In the present study, 230 Koreans (male: 115 and female: 115) and additional 96 Caucasians (male: 50, female: 46) balanced by gender and age (as shown in Table 2) were recruited to the ear measurement survey for use in ear product design for not only Koreans but also Caucasians.

2.2.2. Measurement protocol

The ear dimensions of the right ears were measured using 3D ear scans obtained by scanning and casting the ears. A 3D ear scan was obtained in five phases as shown in Figure 4: (1) introducing the purpose of the

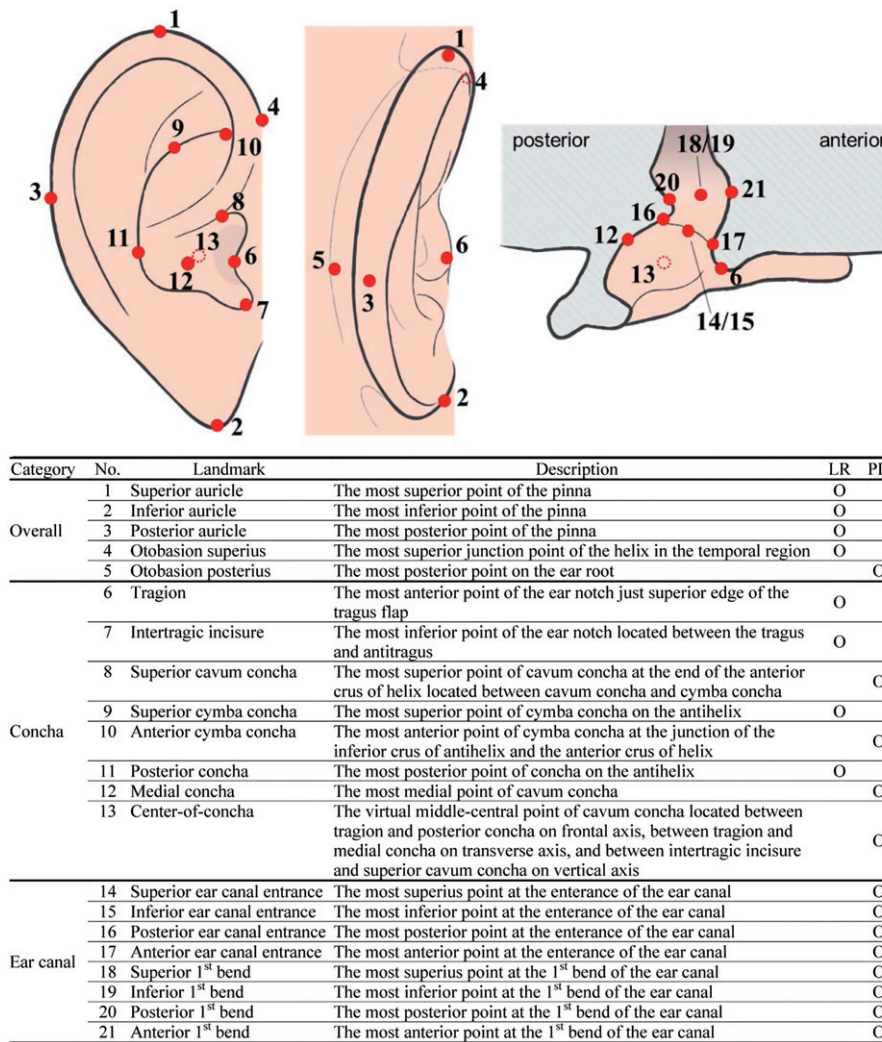


Figure 3. Ear landmarks.

Table 1. Maximum sampling error (SE; unit: mm) and sample size (n) requirement by precision for ear dimensions.

Ear dimensions	1%		2%		3%		4%		5%	
	SE	n	SE	n	SE	n	SE	n	SE	n
Ear length	0.6	423	1.2	106	1.7	47	2.3	26	2.9	17
Ear breadth	0.3	736	0.7	184	1.0	82	1.3	46	1.6	29

study and the process of scanning, (2) scanning the pinna, (3) casting the concha and ear canal, (4) scanning the cast of the concha and ear canal, and (5) merging the scan of the pinna and that of the concha and ear canal cast. In the orientation phase, the purpose and process of ear scanning were explained to the participant, and then informed consent was obtained. In the 3D ear scanning phase, the right pinna was captured using the Artec Eva (Artec Group, Inc., Luxembourg) 3D scanner with Artec Studio 9 (Artec Group, Inc., Luxembourg) for image processing. In the casting phase, the right concha and ear canal

Table 2. Distribution of participants to the ear measurement survey by ethnicity, gender, and age.

Ethnicity	Gender	Age				Sum	
		20s	30s	40s	50s		
Korean	Male	33	26	24	27	115	230
	Female	37	25	27	23	115	
Caucasian	Male	31	13	3	3	50	96
	Female	32	9	1	4	46	
Sum		133	73	55	57	326	

were cast using Otoform (Dreve Otoplastik GmbH, Germany) by referring to the method of crafting custom hearing aid devices (Jung et al. 2014; Mohamed et al. 2013; Paulsen 2004; Sickel et al. 2011). Lastly, the cast of the concha and ear canal was scanned using the Artec Eva scanner and surface of the scanned concha was inverted to generate its original view. The present study measured only the right ears due to the similarity (mean difference <1 mm) between the left and right ears in ear length and ear width (Ahmed

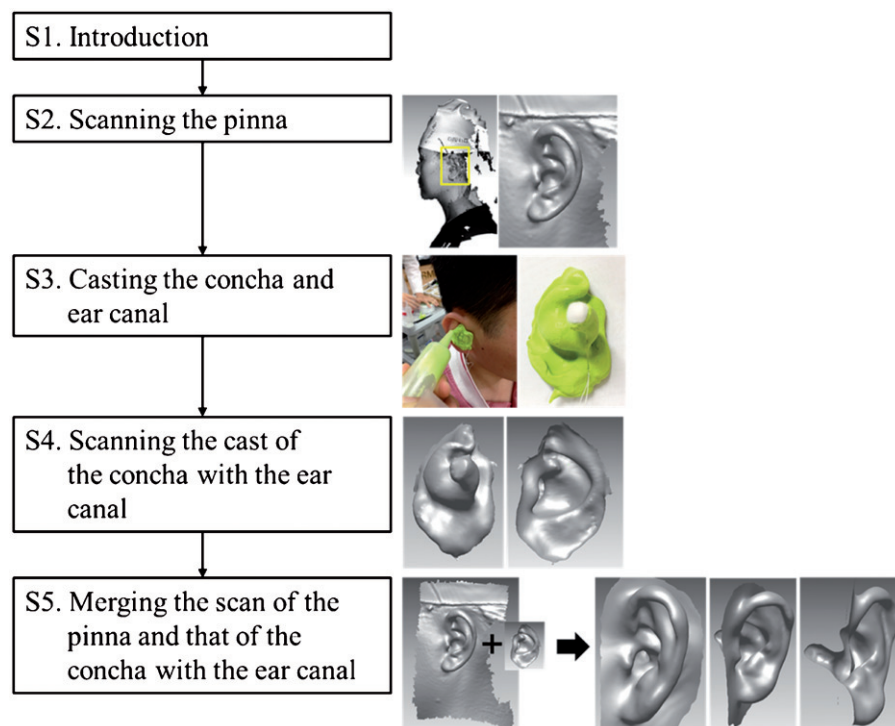


Figure 4. Process of 3D ear scanning for the pinna, concha, and ear canal.

and Omer 2015; Jung and Jung 2003; Taura, Adamu, and Modibbo 2013).

After 3D scanning of the ear, measurement of ear dimensions was performed in three phases: (1) post-processing of the scanned ear, (2) placement of landmarks, and (3) measurement of ear dimensions. As 3D human body scan data commonly have small holes, missing regions, unsmoothed surfaces, erroneous polygons, and noises, post-processing of the scan data is necessary (Allen, Curless, and Popović 2003; Lee et al. 2016b). In the post-processing phase, the scanned pinna and concha with the ear canal were edited by applying the functions of hole-filling (filling small holes or missing regions), smoothing (smoothing bumpy surfaces), and mesh refinement (removing noises or abnormal meshes) provided by Artec Studio 9 and RapidForm 2006 (INUS Technology, Inc., Korea). Then, the 3D scanned pinna was merged with the concha and ear canal into one complete 3D ear using RapidForm 2006. In the landmark placement phase, anthropometric ear landmarks were manually marked by one experimenter on the edited 3D ear image using RapidForm 2006. Lastly, in the measurement phase, ear dimensions were automatically measured by a program developed with Matlab (MathWorks, Inc., Natick, MA, USA) in the present study. Among ear dimensions, length and width dimensions were measured by calculating Euclidian distances among landmarks, and arc dimensions were measured by creating

a virtual plane passing through three landmarks and forming an arc through intersection between the plane and the 3D ear image (Lee et al. 2013a). The integrity of ear dimension measurement was assured by an outlier checking process, in which ear measurements exceeding the range of corresponding mean $\pm 3SD$ were examined for accuracy.

The intra-measure reliability of the ear landmarking process was evaluated in terms of SD and CV of ear measurement. Marking the twenty-one ear landmarks on a digital ear randomly selected from the 3D ear scan database was repeated three times by one experimenter with an interval of at least 24 h. The SD and CV for each of the ear dimensions were found smaller than 2 mm and 5%, respectively, which were used as criteria of satisfactory reliability for hand and head measurement by Lee et al. (2017), Li et al. (2008), and Weinberg et al. (2005).

2.3 Data analysis

T-test and *F*-test were conducted to find out the differences of ear measurements in mean and SD, respectively, based on gender, age, and ethnicity. The ear measurement results of the present study were also compared with the previous nine ear measurement studies (Ahmed and Omer 2015; Alexander and Laubach 1968; Bozkir et al. 2006; Jung and Jung 2003; Liu 2008; Meijerman, van der Lugt, and Maat 2007;

Table 3. Ear anthropometric studies.

No	Ear measurement study	Population	Sample size	Age	No. of dimensions	Measurement method
1	The present study	Korean	Male: 115 Female: 115	20–59	25	3D measurement
		Caucasian	Male: 50 Female: 46			
2	Yoon and Jung (2002)	Korean	Male: 318 Female: 260	18–60	8	Direct measurement
3	Jung and Jung (2003)	Korean	Male: 300 Female: 300	17–89	4	Direct measurement
4	Liu (2008)	Taiwanese	Male: 100 Female: 100	20–59	3	photogrammetry
5	Bozkir et al. (2006)	Turkish	Male: 191 Female: 150	18–25	7	Direct measurement
6	Alexander and Laubach (1968)	Caucasian American (US Air Force)	Male: 500	23.3 ± 6.1	4	photogrammetry
7	Meijerman, van der Lugt, and Maat (2007)	Dutch	Male: 919 Female: 434	18–99	6	photogrammetry
8	Ahmed and Omer (2015)	Sudanese	Male: 100 Female: 100	18–30	9	Direct measurement
9	Taura, Adamu, and Modibbo (2013)	Nigerian	Male: 137 Female: 82	18–25	12	Direct measurement
10	Porter and Olson (2001)	African American	Female: 108	13–30	2	Direct measurement

Porter and Olson 2001; Taura, Adamu, and Modibbo 2013; Yoon and Jung 2002) as shown in Table 3 in terms of ear length, ear breadth, and cavum concha width. Inter-population variabilities (mean differences of the maxima and minima between populations) among the populations and intra-population variabilities (differences of the maxima and minima within a particular population) of Koreans and Caucasians were analysed. Lastly, the Tukey test was conducted for multiple comparisons on the mean differences in ear dimensions between age groups. All statistical tests were conducted at $\alpha = .5$ using Matlab (MathWorks, Inc., USA).

3. Results

3.1. Ear measurements of Koreans and Caucasians

Descriptive statistics (mean, SD, minimum, maximum, and percentiles) of the measured ear dimensions for male and female Koreans and Caucasians are presented in Tables 4, 5, and 6. For example, Table 4 shows the descriptive statistics of ear length (unit: mm) for the male Koreans including mean $\pm$ SD = 64.8 $\pm$ 4.2, minimum = 53.6, maximum = 75.8, $p_{0.01} = 55.4$, $p_{0.05} = 58.2$, $p_{0.95} = 71.8$, and $p_{0.99} = 73.8$.

3.2. Comparison of ear measurements by gender

A comparison in mean and SD between males and females of both Korean and Caucasian participants is presented in Tables 4 and 5, which reveal that males have a slightly bigger ear and more varied than female in most of non-angle dimensions. Korean males were found significantly larger than Korean females in all non-angle dimensions ($\bar{d} = 0.5$ –3.5 mm; ratio of means = 1.03–1.18). Caucasian males also showed similar or larger sizes than Caucasian females ($\bar{d} = 0.0$ –2.8 mm; ratio of means = 1.00–1.13) in most non-angle dimensions, except cavum concha length

($\bar{d} = -0.3$ mm) with significant difference in nine dimensions. Results of the SD ratio analysis indicate that the ear measurements of both Korean and Caucasian males were similar (ratio of SDs $\cong 1$) or more dispersed (ratio of SDs > 1.1) than those of Korean and Caucasian females in most non-angle dimensions (ratio of SDs = 1.01–1.51), but significant differences in SD were rarely found. Lastly, angle dimensions indicate that Korean males and females showed significant differences in mean and SD for ear angle ($\bar{d} = -0.7^\circ$; ratio in mean = 0.87), pinna flare angle ($\bar{d} = 2.4^\circ$; ratio in mean = 1.09), and ear canal azimuth angle ($\bar{d} = 6.1^\circ$; ratio in mean = 1.04; ratio of SD = 0.76), while Caucasians had no significant difference in mean difference of angle dimensions ($\bar{d} = -3.1$ –2.5 $^\circ$) by gender, but Caucasian males showed less variations (ratio of SDs < 0.9) in ear angle dimensions than females (ratio of SDs = 0.79–0.89).

3.3. Comparison of ear measurements between Koreans and Caucasians

A comparison in mean and SD between Korean and Caucasian gender-composite populations presented in Table 6 indicates that Koreans tend to have a significantly longer but narrower ear and more varied than Caucasians. Koreans were found to be significantly larger than Caucasians in most of the non-angle dimensions ($\bar{d} = 0.4$ –3.7 mm; ratio of means = 1.02–1.15), except ear breadth ($\bar{d} = -1.4$ mm; ratio of means = 0.96). Results of the SD ratio analysis show that the ear measurements of Koreans were similar (ratio of SDs $\cong 1$) or more dispersed (ratio of SDs > 1.1) than those of Caucasians in most ear dimensions (ratio of SDs = 1.01–1.55) except otobasion superius to otobasion posterius vertical length (ratio of SD = 0.78), center-of-concha to otobasion posterius length (ratio of SD = 0.95), center-of-concha to inter-tragic incisure length (ratio of SD = 0.98), and ear canal entrance height (ratio of SD = 0.94), and the

Table 4. Comparison of ear measurements between Korean males and females (unit: mm, °).

No.	Ear dimensions	Male													Female													Male (m) vs. Female (f)		
		n	Percentile												n	Percentile												Mean Difference ($M_m - M_f$)	Mean Ratio ($\frac{M_m}{M_f}$)	SD Ratio ($\frac{SD_m}{SD_f}$)
			Mean	SD	Min	Max	1st	5th	10th	25th	75th	90th	95th	99th		Mean	SD	Min	Max	1st	5th	10th	25th	75th	90th	95th	99th			
1	Ear length	115	64.8	4.2	53.6	75.8	55.4	58.2	59.6	62.6	67.2	70.9	71.8	73.8	115	61.8	4.0	52.8	70.3	53.1	54.7	56.4	59.4	64.9	66.9	68.0	69.3	3.0**	1.05	1.05
2	Otobasion superius to otobasion posterius horizontal length	115	24.6	2.3	18.1	31.7	18.6	20.7	21.8	23.1	26.0	27.1	28.0	29.1	115	22.6	2.0	16.0	27.4	18.1	19.9	20.2	21.2	24.1	25.1	25.9	27.2	2.0**	1.09	1.12
3	Otobasion superius to otobasion posterius vertical length	115	23.7	2.8	17.1	30.9	17.3	19.1	20.1	21.9	25.9	27.3	28.0	30.2	115	23.1	3.0	14.5	31.8	16.7	17.6	19.7	21.1	24.8	26.8	28.0	29.6	0.6	1.03	0.93
4	Center-of-concha to otobasion posterius length	115	31.4	3.1	25.2	40.1	25.5	27.0	27.6	29.0	33.4	35.5	37.0	39.1	115	29.6	2.9	22.2	37.7	22.7	24.4	25.8	27.5	31.6	33.0	33.5	36.4	1.9**	1.06	1.06
5	Center-of-concha to otobasion superius length	115	31.7	2.5	25.1	37.3	26.3	27.4	28.7	30.1	33.4	34.7	35.6	37.1	115	30.4	2.1	25.5	35.9	25.7	27.0	27.6	29.0	31.8	33.1	34.0	34.7	1.3**	1.04	1.16
6	Ear breadth	115	33.2	2.4	28.6	41.0	28.7	29.4	29.7	31.5	34.9	35.9	37.0	39.8	115	29.7	2.4	24.0	35.9	24.4	25.7	27.0	28.2	31.2	32.7	33.8	34.9	3.5**	1.12	1.01
7	Ear protrusion	115	17.5	2.9	11.0	26.9	11.2	13.1	14.2	15.6	19.5	21.1	22.8	23.9	115	15.1	2.6	9.1	25.2	10.9	11.3	11.9	13.2	16.5	18.3	19.4	21.1	2.4*	1.16	1.10
8	Upper otobasion arc	115	48.8	6.7	39.0	77.3	39.9	41.6	43.0	44.5	50.0	59.2	63.3	68.1	115	45.6	5.5	35.4	67.1	37.3	39.5	39.9	41.9	48.0	52.8	57.5	63.3	3.2**	1.07	1.21*
9	Ear angle	115	4.6	2.3	-7.5	11.0	-0.5	1.0	1.9	3.3	5.6	7.4	7.6	8.8	115	5.2	2.1	-1.1	12.7	-0.1	1.8	3.0	4.6	6.1	7.8	8.4	11.2	-0.7*	0.87	1.09
10	Pinna flare angle	115	30.1	7.5	14.3	55.4	15.5	20.1	21.4	25.3	34.3	39.1	43.9	53.5	115	27.7	7.4	17.1	52.3	17.1	18.7	20.3	22.4	31.7	37.2	40.8	48.4	2.4**	1.09	1.01
11	Cavum concha length	115	17.2	1.6	11.3	22.2	13.6	14.4	15.3	16.2	18.2	19.1	19.7	21.2	115	16.7	1.5	13.6	21.1	13.7	14.6	15.0	15.6	17.6	18.6	19.0	20.7	0.5*	1.03	1.11
12	Center-of-concha to anterior cymba concha length	115	7.8	1.4	4.7	11.6	5.1	5.6	6.1	6.9	8.5	9.4	10.1	11.5	115	7.3	1.4	4.6	12.6	4.9	5.5	5.8	6.4	7.8	9.1	10.1	11.9	0.5**	1.07	0.95
13	Center-of-concha to superior cymba concha length	115	17.2	1.4	14.4	22.0	14.5	15.4	15.6	16.3	17.9	18.9	20.2	21.2	115	16.6	1.7	12.6	21.6	12.6	14.3	14.8	15.6	17.3	18.9	19.8	20.5	0.7**	1.04	0.86
14	Superior cavum concha to anterior cymba concha length	115	6.5	1.6	3.1	9.9	3.1	3.9	4.4	5.4	7.7	8.4	9.1	9.6	115	5.5	1.6	1.6	9.1	2.2	2.8	3.4	4.2	6.8	7.5	7.8	9.0	1.0**	1.18	0.97
15	Posterior concha to anterior cymba concha length	115	16.5	2.1	11.1	20.8	11.9	12.8	13.4	15.3	17.8	19.3	19.9	20.6	115	15.4	2.1	10.5	21.7	11.2	12.6	13.0	14.1	16.7	18.0	18.8	21.1	1.0**	1.07	1.01
																	(continued)													

(continued)

Table 4. Continued.

No.	Ear dimensions	Male														Female														Male (m) vs. Female (f)		
		Percentile														Percentile														Mean Difference ($M_m - M_f$)	Mean Ratio ($\frac{M_m}{M_f}$)	SD Ratio ($\frac{SD_m}{SD_f}$)
		n	Mean	SD	Min	Max	1st	5th	10th	25th	75th	90th	95th	99th	n	Mean	SD	Min	Max	1st	5th	10th	25th	75th	90th	95th	99th					
16	Center-of-concha to intertragic incisive length	115	10.9	1.4	7.8	14.4	7.9	9.0	9.2	9.9	11.8	12.6	13.2	13.8	115	10.0	1.3	7.4	13.8	7.7	8.1	8.4	9.1	10.9	11.6	12.2	13.4	0.9**	1.09	1.06		
17	Cavum concha width	115	17.4	1.9	12.7	22.0	12.9	14.1	14.7	16.4	18.9	19.5	20.4	20.8	115	16.3	1.8	11.3	21.2	12.3	13.5	14.3	15.2	17.4	18.6	19.4	20.6	1.1**	1.07	1.02		
18	Cavum concha depth	115	7.9	1.2	5.5	11.8	5.6	6.1	6.5	7.1	8.7	9.5	10.1	11.6	115	7.1	1.2	4.3	10.8	4.6	5.2	5.8	6.2	7.8	8.4	8.9	10.4	0.9**	1.12	1.07		
19	Ear canal entrance height	115	14.2	1.3	11.4	17.5	11.7	12.0	12.6	13.2	15.1	15.7	16.3	16.5	115	13.4	1.3	9.8	16.5	10.4	11.0	11.9	12.5	14.2	15.1	15.5	16.0	0.8**	1.06	0.98		
20	Ear canal entrance width	115	7.9	1.2	5.6	10.9	5.8	6.2	6.5	7.1	8.7	9.5	10.2	10.5	115	7.5	1.0	5.0	10.7	5.3	5.9	6.2	6.7	8.2	8.7	9.0	10.3	0.4**	1.06	1.10		
21	Ear canal entrance to 1 st bend length	115	4.2	0.6	2.5	5.7	2.8	3.2	3.4	3.7	4.7	5.0	5.2	5.5	115	3.9	0.6	2.4	5.8	2.5	2.9	3.1	3.5	4.3	4.6	4.8	5.7	0.3**	1.09	1.01		
22	Ear canal entrance circumference	115	37.4	3.0	29.8	44.4	31.8	32.7	33.6	35.8	39.3	41.2	42.2	44.3	115	35.3	2.8	28.1	42.6	29.3	30.8	31.3	33.9	37.3	38.9	39.6	41.8	2.1**	1.06	1.06		
23	Ear canal 1 st bend circumference	115	29.5	3.0	19.7	36.1	21.9	25.0	25.7	27.5	31.5	33.2	33.7	35.7	115	27.9	3.5	18.0	34.7	19.0	22.1	23.3	26.0	30.2	32.4	33.8	34.2	1.6**	1.06	0.85		
24	Ear canal azimuth angle	115	161.7	15.8	104.0	180.0	114.4	130.2	141.3	154.1	173.6	178.6	178.8	179.9	115	155.6	20.9	79.2	179.9	87.5	114.0	125.0	148.1	170.3	176.7	178.6	179.8	6.1*	1.04	0.76**		
25	Ear canal elevation angle	115	96.6	10.8	75.2	131.6	79.4	80.5	83.9	88.8	102.2	111.4	115.6	123.4	115	99.1	10.4	74.6	126.6	77.4	82.3	85.0	92.7	105.1	112.8	117.5	122.8	-2.5	0.97	1.04		

 * $p < 0.05$, ** $p < 0.01$.

Table 5. Comparison of ear measurements between Caucasian males and females (unit: mm, °).

No.	Ear dimensions	Male													Female													Male (m) vs. Female (f)		
		Percentile													Percentile															
		N	Mean	SD	Min	Max	1st ^t	5th	10th	25th	75th	90th th	95th	99th	n	Mean	SD	Min	Max	1st	5th	10th	25th	75th	90th th	95th	99th	Mean Difference ($M_m - M_f$)	Mean Ratio ($\frac{M_m}{M_f}$)	SD Ratio ($\frac{SD_m}{SD_f}$)
1	Ear length	50	61.6	3.8	52.0	68.6	53.8	56.9	57.6	58.7	64.1	67.0	67.2	68.2	46	59.0	4.3	48.7	70.8	50.1	52.7	53.6	56.4	60.6	63.5	65.0	70.7	2.6**	1.04	0.87
2	Otobasion superius to otobasion posterius horizontal length	50	23.4	2.0	19.2	28.0	19.2	19.9	21.4	22.0	24.6	26.3	26.9	27.9	46	22.6	1.9	19.1	27.6	19.2	19.7	20.4	21.3	23.6	24.8	25.5	27.3	0.8*	1.04	1.09
3	Otobasion superius to otobasion posterius vertical length	50	21.8	3.7	14.1	31.6	14.8	16.3	17.2	18.7	23.8	25.8	28.0	31.3	46	21.8	3.8	13.8	32.3	14.9	16.2	16.6	19.3	23.9	26.9	27.3	30.2	0.0	1.00	0.98
4	Center-of-concha to otobasion posterius length	50	29.1	3.4	20.1	35.8	21.0	23.4	25.2	26.9	31.6	33.3	34.1	35.5	46	27.7	3.1	21.0	36.0	21.6	22.7	23.8	25.8	29.8	31.2	32.3	34.6	1.4*	1.05	1.09
5	Center-of-concha to otobasion superius length	50	30.8	2.3	26.2	36.3	26.7	27.6	27.9	29.2	32.3	33.5	34.3	35.8	46	30.0	1.8	25.2	33.4	25.4	26.5	27.7	29.0	31.2	32.1	32.3	32.9	0.8	1.03	1.23
6	Ear breadth	50	34.2	2.5	29.4	40.0	29.4	30.1	31.3	32.5	36.1	37.7	38.4	39.3	46	31.4	2.6	24.2	37.8	25.8	27.9	28.7	29.9	32.6	34.9	36.4	37.6	2.8**	1.09	0.96
7	Ear protrusion	50	16.7	3.6	10.5	25.9	10.9	11.9	12.0	14.4	18.7	21.3	23.5	25.4	46	16.4	2.4	10.7	21.2	10.9	11.7	12.9	15.1	17.6	19.2	20.2	20.9	0.3	1.02	1.51**
8	Upper otobasion arc	50	43.9	3.9	37.1	54.0	37.9	39.0	40.0	41.1	46.5	49.6	51.9	53.3	46	43.1	4.3	34.7	56.5	34.8	35.8	38.2	40.8	44.7	47.7	51.5	54.9	0.8	1.02	0.89
9	Ear angle	50	4.0	2.0	0.1	8.8	0.2	0.9	1.6	2.8	5.7	6.5	7.2	8.3	46	4.2	2.3	0.6	9.8	0.8	1.2	1.7	2.2	5.6	7.1	8.9	9.6	-0.1	0.96	0.87
10	Pinna flare angle	50	28.4	6.2	16.9	52.8	17.5	19.9	22.8	24.7	31.1	35.3	37.8	46.9	46	28.0	7.6	11.2	46.7	14.1	18.3	19.2	22.8	31.4	36.9	43.5	45.8	0.4	1.01	0.82
11	Cavum concha length	50	14.7	1.6	11.7	17.6	11.7	11.9	12.4	13.6	15.9	16.9	17.4	17.5	46	15.0	1.5	12.0	18.4	12.2	12.6	12.9	14.1	15.9	16.8	17.2	18.3	-0.3	0.98	1.11
12	Center-of-concha to anterior cymba concha length	50	7.4	1.0	5.1	9.8	5.3	5.7	6.1	6.9	7.9	8.4	8.6	9.5	46	6.9	0.8	5.5	8.7	5.6	5.8	5.9	6.3	7.5	8.0	8.0	8.5	0.5**	1.07	1.22
13	Center-of-concha to superior cymba concha length	50	16.2	1.4	12.0	19.2	12.8	13.8	14.8	15.4	17.0	18.0	18.9	19.2	46	15.8	1.3	12.2	18.4	12.9	13.8	14.0	15.0	16.8	17.4	18.1	18.3	0.4	1.02	1.08
14	Superior cavum concha to anterior cymba concha length	50	6.7	1.5	3.5	10.2	4.1	4.9	5.1	5.6	7.8	8.7	9.1	10.1	46	6.0	1.3	3.2	9.6	3.5	4.4	4.5	5.3	6.7	7.9	8.1	9.0	0.7*	1.11	1.15
15	Posterior concha to anterior cymba concha length	50	15.8	1.9	11.2	20.0	11.6	13.1	13.2	14.8	17.2	18.0	18.6	19.5	46	14.9	1.4	12.0	18.5	12.0	13.0	13.3	13.8	15.8	16.6	16.9	18.2	0.9*	1.06	1.31
16	Center-of-concha to intertragic incisure length	50	10.9	1.5	7.9	14.8	8.0	8.3	9.2	10.1	11.7	12.7	13.5	14.3	46	10.6	1.3	7.2	14.1	7.4	8.2	9.3	9.8	11.5	12.1	12.6	13.5	0.3	1.03	1.12

(continued)

Table 5. Continued.

No.	Ear dimensions	Male													Female													Male (m) vs. Female (f)		
		Percentile													Percentile													Mean Difference ($M_m - M_f$)	Mean Ratio ($\frac{M_m}{M_f}$)	SD Ratio ($\frac{SD_m}{SD_f}$)
		N	Mean	SD	Min	Max	1st st	5th	10th	25th	75th	90th	95th	99th	n	Mean	SD	Min	Max	1st	5th	10th	25th	75th	90th th	95th	99th			
17	Cavum concha width	50	16.8	1.9	12.2	20.5	12.5	13.9	14.0	15.7	18.4	18.9	19.9	20.5	46	16.0	1.5	12.2	19.7	12.5	13.4	14.4	15.0	17.0	17.6	18.0	19.4	0.8*	1.05	1.27
18	Cavum concha depth	50	6.9	1.1	4.5	9.6	4.7	5.0	5.5	6.1	7.6	8.4	8.7	9.2	46	6.4	0.9	4.7	8.9	4.9	5.3	5.4	5.8	6.9	7.5	7.9	8.7	0.4	1.06	1.28
19	Ear canal entrance height	50	13.0	1.4	10.7	15.9	10.7	10.8	11.1	12.0	13.9	15.0	15.4	15.8	46	12.7	1.4	9.2	15.1	9.3	10.4	11.4	11.9	13.6	14.6	14.9	15.1	0.4	1.03	1.06
20	Ear canal entrance width	50	7.1	1.0	4.7	9.2	4.8	5.5	5.7	6.4	7.9	8.3	8.6	9.1	46	6.3	1.0	4.3	8.5	4.3	5.0	5.2	5.8	6.8	7.7	8.1	8.4	0.8**	1.13	1.07
21	Ear canal entrance to 1 st bend length	50	3.8	0.8	2.4	5.9	2.4	2.5	2.7	3.2	4.2	4.8	4.9	5.6	46	3.6	0.8	2.2	5.3	2.2	2.5	2.6	2.9	4.0	4.8	4.9	5.2	0.2	1.06	1.00
22	Ear canal entrance circumference	50	34.5	3.1	28.1	41.8	28.9	29.9	30.2	32.6	36.2	38.4	39.6	41.5	46	33.1	2.6	27.2	37.7	27.6	29.2	29.6	31.3	35.1	36.5	37.2	37.5	1.4*	1.04	1.18
23	Ear canal 1st bend circumference	50	27.2	3.2	21.0	35.8	21.5	22.5	23.3	24.5	29.3	30.3	32.3	34.8	46	26.6	3.4	17.7	32.9	18.7	22.1	22.9	24.2	28.8	31.5	32.1	32.7	0.6	1.02	0.94
24	Ear canal azimuth angle	50	165.5	10.9	135.4	179.7	136.4	147.7	149.7	159.5	173.8	177.6	178.8	179.4	46	163.0	12.2	124.8	179.9	128.7	138.4	148.6	158.0	170.4	176.6	177.8	179.9	2.5	1.02	0.89
25	Ear canal elevation angle	50	93.1	9.5	72.6	114.9	75.1	79.4	81.0	86.3	99.9	103.8	108.3	112.4	46	96.2	12.0	69.4	125.0	73.1	79.1	84.0	89.1	102.8	114.4	118.5	123.5	-3.1	0.97	0.79

 * $p < 0.05$, ** $p < 0.01$.

Table 6. Comparison of ear measurements between Koreans and Caucasians (unit: mm, °).

		Korean														Caucasian														Korean (K) vs. Caucasian (C)		
No.	Ear dimensions	n	Mean	SD	Min	Max	1st	5th	10th	Percentile							Mean	Difference (M _K - M _C)	Ratio ($\frac{M_K}{M_C}$)	SD Ratio ($\frac{SD_K}{SD_C}$)												
										25th	75th	90th	95th	99th	n	Mean					SD	Min	Max	1st	5th	10th	25th	75th	90th	95th	99th	
1	Ear length	230	63.3	4.3	52.8	75.8	53.7	55.6	57.9	60.4	66.2	68.5	70.8	73.0	96	60.4	4.2	48.7	70.8	51.6	53.5	55.3	57.9	63.5	65.8	67.2	70.5	2.9**	1.05	1.02		
2	Otobasion superius to otobasion posterius horizontal length	230	23.6	2.4	16.0	31.7	18.2	20.0	20.5	21.9	25.2	26.7	27.2	28.8	96	23.0	2.0	19.1	28.0	19.2	19.8	20.5	21.7	24.2	25.5	26.8	27.7	0.6*	1.02	1.19*		
3	Otobasion superius to otobasion posterius vertical length	230	23.4	2.9	14.5	31.8	16.9	18.6	19.8	21.4	25.5	27.0	28.0	30.2	96	21.8	3.7	13.8	32.3	14.1	16.2	16.7	19.1	23.8	26.6	27.5	31.7	1.6**	1.07	0.78**		
4	Center-of-concha to otobasion posterius length	230	30.5	3.2	22.2	40.1	23.5	25.6	26.7	28.4	32.6	34.2	36.3	37.8	96	28.4	3.3	20.1	36.0	20.9	22.7	24.3	26.1	30.9	32.4	33.5	35.8	2.1**	1.07	0.95		
5	Center-of-concha to otobasion superius length	230	31.0	2.4	25.1	37.3	25.8	27.1	28.1	29.3	32.7	34.1	34.8	37.0	96	30.4	2.1	25.2	36.3	25.7	27.2	27.8	29.1	31.7	33.0	33.6	35.2	0.6*	1.02	1.14		
6	Ear breadth	230	31.4	3.0	24.0	41.0	24.8	27.0	27.8	29.3	33.7	35.0	36.0	37.4	96	32.9	2.9	24.2	40.0	27.7	28.8	29.4	31.0	35.0	36.7	37.9	38.7	-1.4**	0.96	1.02		
7	Ear protrusion	230	16.3	3.0	9.1	26.9	11.0	11.6	12.6	14.2	18.2	20.2	21.4	23.8	96	16.6	3.0	10.5	25.9	10.7	11.7	12.5	14.9	18.1	20.3	21.3	24.8	-0.2	0.98	1.00		
8	Upper otobasion arc	230	47.2	6.3	35.4	77.3	38.8	39.8	40.7	43.4	49.0	56.5	61.4	67.2	96	43.5	4.1	34.7	56.5	34.9	37.7	39.4	40.9	45.6	48.7	52.0	54.1	3.7**	1.08	1.55**		
9	Ear angle	230	4.9	2.2	-7.5	12.7	-0.6	1.3	2.3	3.8	5.9	7.5	8.1	11.0	96	4.1	2.2	0.1	9.8	0.4	1.1	1.6	2.5	5.7	6.9	7.9	9.3	0.8**	1.19	1.02		
10	Pinna flare angle	230	28.9	7.5	14.3	55.4	17.1	18.9	20.7	23.6	33.3	38.2	42.6	51.9	96	28.2	6.9	11.2	52.8	16.6	18.6	20.2	23.8	31.4	35.6	41.3	47.0	0.7	1.03	1.09		
11	Cavum concha length	230	16.9	1.6	11.3	22.2	13.6	14.5	15.1	15.9	18.0	18.9	19.6	21.0	96	14.8	1.6	11.7	18.4	11.7	12.2	12.6	13.8	15.9	16.9	17.4	18.2	2.1**	1.14	1.01		
12	Center-of-concha to anterior cymba concha length	230	7.5	1.4	4.6	12.6	4.9	5.6	5.9	6.6	8.2	9.3	10.2	11.6	96	7.2	0.9	5.1	9.8	5.5	5.8	5.9	6.4	7.8	8.3	8.5	9.3	0.4*	1.05	1.55**		
13	Center-of-concha to superior cymba concha length	230	16.9	1.6	12.6	22.0	12.7	14.7	15.2	15.9	17.7	19.0	20.1	21.2	96	16.0	1.4	12.0	19.2	12.2	13.8	14.4	15.2	17.0	17.6	18.5	19.1	0.9**	1.06	1.14		
14	Superior cavum concha to anterior cymba concha length	230	6.0	1.7	1.6	9.9	2.3	3.3	3.8	4.7	7.3	8.0	8.5	9.5	96	6.4	1.4	3.2	10.2	3.5	4.4	4.8	5.4	7.2	8.4	9.0	10.0	-0.4	0.94	1.17		
15	Posterior concha to anterior cymba concha length	230	16.0	2.1	10.5	21.7	11.2	12.6	13.1	14.5	17.5	18.6	19.5	20.8	96	15.4	1.7	11.2	20.0	11.9	13.0	13.2	14.1	16.3	17.5	18.2	19.0	0.6*	1.04	1.26*		
16	Center-of-concha to intertragic incisure length	230	10.5	1.4	7.4	14.4	7.7	8.3	8.7	9.5	11.4	12.3	12.9	13.8	96	10.8	1.4	7.2	14.8	7.7	8.2	9.2	9.9	11.5	12.6	13.0	14.1	-0.3	0.97	0.98		
17	Cavum concha width	230	16.8	1.9	11.3	22.0	12.5	13.8	14.4	15.5	18.2	19.3	19.9	20.8	96	16.4	1.8	12.2	20.5	12.2	13.4	14.2	15.2	17.5	18.7	19.3	20.4	0.5*	1.03	1.08		
18	Cavum concha depth	230	7.5	1.3	4.3	11.8	4.7	5.6	5.9	6.6	8.3	9.1	9.7	10.8	96	6.7	1.0	4.5	9.6	4.7	5.1	5.4	5.9	7.3	8.0	8.5	8.9	0.8**	1.13	1.24*		
19	Ear canal entrance height	230	13.8	1.3	9.8	17.5	10.6	11.7	12.1	13.0	14.9	15.5	15.8	16.5	96	12.9	1.4	9.2	15.9	9.5	10.7	11.1	11.9	13.8	14.8	15.1	15.6	0.9**	1.07	0.94		
20	Ear canal entrance width	230	7.7	1.1	5.0	10.9	5.6	6.0	6.3	6.9	8.4	9.1	9.6	10.5	96	6.7	1.1	4.3	9.2	4.3	5.0	5.5	5.9	7.5	8.1	8.4	9.0	1.0**	1.15	1.04		
21	Ear canal entrance to 1 st bend length	230	4.1	0.7	2.4	5.8	2.6	2.9	3.2	3.6	4.5	4.9	5.2	5.6	96	3.7	0.8	2.2	5.9	2.3	2.5	2.7	3.1	4.2	4.8	4.9	5.4	0.4**	1.10	0.83*		
22	Ear canal entrance circumference	230	36.4	3.1	28.1	44.4	29.7	31.3	32.2	34.4	38.2	40.3	41.4	44.2	96	33.8	2.9	27.2	41.8	28.0	29.4	30.0	31.8	35.7	37.3	38.4	41.3	2.6**	1.08	1.04		
23	Ear canal 1 st bend circumference	230	28.7	3.3	18.0	36.1	19.6	22.8	24.4	26.8	31.1	33.0	33.8	34.6	96	26.9	3.3	17.7	35.8	19.8	22.2	23.1	24.4	29.1	31.2	32.3	33.8	1.8**	1.07	1.00		
24	Ear canal azimuth angle	230	158.7	18.7	79.2	180.0	92.2	122.4	132.8	150.8	172.1	178.1	178.8	179.9	96	164.3	11.6	124.8	179.9	133.0	141.4	148.9	158.5	172.4	176.9	178.7	179.9	-5.6**	0.97	1.62**		
25	Ear canal elevation angle	230	97.8	10.6	74.6	131.6	77.7	81.6	84.6	90.3	103.7	112.0	116.7	123.4	96	94.6	10.8	69.4	125.0	72.4	78.8	81.7	88.0	101.2	108.4	115.1	121.9	3.3*	1.03	0.99		

* $p < 0.05$, ** $p < 0.01$.

significance of differences in SD are revealed in 8 of 25 ear dimensions. Lastly, angle dimensions indicate that significant differences between Koreans and Caucasians in mean for ear angle ($\bar{d}=0.8^\circ$; ratio in mean = 1.19), ear canal azimuth angle ($\bar{d}=-5.6^\circ$; ratio in mean = 0.97), and ear canal elevation angle ($\bar{d}=3.3^\circ$; ratio in mean = 1.03).

3.4. Comparison of ear measurements by age

Tukey test results on mean differences of ear measurements among age groups (see Table 7) indicate a trend that the height (vertical length) of pinna increases but the width (horizontal length) of the concha decreases by age. A significant increase in mean at $\alpha=.05$, when people in their 50s were compared with those in their 20s, was found in three pinna-height related dimensions including ear length (ratio of mean ear length of the age group 50s to that of the age group 20s = 1.06), otobasion superior to otobasion posterior vertical length (1.08), and center-of-concha to otobasion superius length (1.05). In contrast, the opposite was observed in four concha-width related dimensions including center-of-concha to anterior cymba concha length (.89), posterior concha to anterior cymba concha length (.92), center-of-concha to intertragic incisure length (.92), and cavum concha width (.96).

3.5. Comparison of ear measurements between ear surveys

Measurements of ear dimensions were compared with the existing ear anthropometric studies. Three ear dimensions (ear length, EL; ear breadth, EB; and cavum concha width, CCW) measured in common in the previous studies were compared with those in the present study as shown in Table 8. Inter-population variabilities of the three ear dimensions by gender ($\Delta\bar{EL}_{\text{male}}=12.6\text{ mm}$, $\Delta\bar{EL}_{\text{female}}=10.1\text{ mm}$, $\Delta\bar{EB}_{\text{male}}=4.1\text{ mm}$, $\Delta\bar{EB}_{\text{female}}=3.3\text{ mm}$, $\Delta\bar{CCW}_{\text{male}}=1.9\text{ mm}$, and $\Delta\bar{CCW}_{\text{female}}=1.6\text{ mm}$) are smaller than the intra-population variabilities of Koreans by gender ($\Delta EL_{\text{male}}=22.2\text{ mm}$, $\Delta EL_{\text{female}}=17.5\text{ mm}$, $\Delta EB_{\text{male}}=12.4\text{ mm}$, $\Delta EB_{\text{female}}=11.9\text{ mm}$, $\Delta CCW_{\text{male}}=9.3\text{ mm}$, and $\Delta CCW_{\text{female}}=9.9\text{ mm}$) and Caucasians by gender ($\Delta EL_{\text{male}}=16.6\text{ mm}$, $\Delta EL_{\text{female}}=22.1\text{ mm}$, $\Delta EB_{\text{male}}=10.6\text{ mm}$, $\Delta EB_{\text{female}}=13.6\text{ mm}$, $\Delta CCW_{\text{male}}=8.3\text{ mm}$, and $\Delta CCW_{\text{female}}=7.5\text{ mm}$). The mean differences between gender for populations (range of $\Delta\bar{EL}=1.3\text{--}7.0\text{ mm}$, range of $\Delta\bar{EB}=1.1\text{--}3.5\text{ mm}$, and range of $\Delta\bar{CCW}=0.7\text{--}1.1\text{ mm}$) also are smaller than the minimum value of intra-population variability ($\Delta CCW_{\text{female}}=7.5\text{ mm}$). Lastly, males were found having significantly larger ears on average than females in most populations (ear length: $\bar{d}=1.3\text{--}7.0\text{ mm}$; ear breadth: $\bar{d}=1.1\text{--}3.5\text{ mm}$; cavum concha width: $\bar{d}=0.7\text{--}1.1\text{ mm}$).

Table 7. Comparison of ear measurements between age groups 20s–50s.

No.	Ear dimensions	20s (n = 141)			30s (n = 73)			40s (n = 55)			50s (n = 57)		
		mean	SD	ratio	mean	SD	ratio	mean	SD	ratio	mean	SD	ratio
1	Ear length	61.2 ^a	4.2	1.00	62.1 ^{a,b}	4.6	1.01	63.7 ^b	4.4	1.04	64.7 ^b	4.1	1.06
2	Otobasion superius to otobasion posterius horizontal length	23.3	2.1	1.00	23.4	2.2	1.00	23.4	2.4	1.00	23.7	2.7	1.02
3	Otobasion superius to otobasion posterius vertical length	22.3 ^a	3.2	1.00	22.9 ^{a,b}	3.4	1.03	23.3 ^{a,b}	2.7	1.04	24.2 ^b	3.2	1.08
4	Center-of-concha to otobasion posterius length	29.2 ^a	3.2	1.00	30.0 ^{a,b}	3.7	1.03	30.6 ^b	3.1	1.05	30.6 ^b	3.1	1.05
5	Center-of-concha to otobasion superius length	31.0	2.2	1.00	30.8	2.3	0.99	30.5	2.3	0.98	30.9	2.6	1.00
6	Ear breadth	32.2	2.7	1.00	31.5	3.2	0.98	31.5	3.1	0.98	31.8	3.3	0.99
7	Ear protrusion	16.1	2.5	1.00	16.5	2.9	1.03	16.1	3.2	1.01	17.3	3.8	1.08
8	Upper otobasion arc	46.3	7.5	1.00	45.7	5.5	0.99	46.2	3.9	1.00	45.8	3.8	0.99
9	Ear angle	4.5	2.6	1.00	4.4	2.0	0.97	5.0	2.0	1.11	5.0	1.7	1.10
10	Pinna flare angle	28.3	6.7	1.00	28.3	7.0	1.00	28.0	6.8	0.99	30.9	9.2	1.09
11	Cavum concha length	16.2	2.0	1.00	16.1	1.8	1.00	16.6	1.7	1.03	16.7	1.6	1.03
12	Center-of-concha to anterior cymba concha length	7.8 ^a	1.4	1.00	7.4 ^{a,b}	1.1	0.95	7.1 ^b	1.4	0.91	6.9 ^b	1.0	0.89
13	Center-of-concha to superior cymba concha length	16.7	1.7	1.00	16.7	1.6	1.00	16.5	1.5	0.99	16.5	1.3	0.99
14	Superior cavum concha to anterior cymba concha length	6.3	1.5	1.00	6.3	1.7	1.01	5.8	1.5	0.91	5.7	1.8	0.91
15	Posterior concha to anterior cymba concha length	16.3 ^a	2.0	1.00	15.8 ^{a,b}	1.9	0.97	15.2 ^b	2.1	0.93	15.1 ^b	1.9	0.92
16	Center-of-concha to intertragic incisure length	11.1 ^a	1.4	1.00	10.3 ^b	1.3	0.93	10.0 ^b	1.4	0.91	10.2 ^b	1.2	0.92
17	Cavum concha width	17.1 ^a	1.8	1.00	16.7 ^{a,b}	1.9	0.98	16.1 ^b	1.9	0.95	16.3 ^b	1.9	0.96
18	Cavum concha depth	7.4	1.4	1.00	7.1	1.3	0.95	7.0	0.9	0.94	7.4	1.2	0.99
19	Ear canal entrance height	13.4	1.5	1.00	13.5	1.4	1.01	13.8	1.3	1.03	13.7	1.4	1.02
20	Ear canal entrance width	7.3	1.3	1.00	7.5	1.1	1.01	7.3	1.1	1.00	7.5	1.2	1.02
21	Ear canal entrance to 1st bend length	3.9	0.8	1.00	4.0	0.7	1.03	4.0	0.6	1.04	3.9	0.8	1.00
22	Ear canal entrance circumference	35.4	3.4	1.00	35.8	3.1	1.01	35.9	3.2	1.02	35.8	3.1	1.01
23	Ear canal 1st bend circumference	28.1	3.2	1.00	27.9	3.3	0.99	28.8	3.0	1.02	28.2	4.3	1.00
24	Ear canal azimuth angle	159.9	18.4	1.00	160.0	16.2	1.00	160.9	16.4	1.01	161.4	15.9	1.01
25	Ear canal elevation angle	97.5	11.8	1.00	97.5	10.6	1.00	95.2	10.4	0.98	96.1	8.5	0.99

Note: The superscripts a and b on mean values indicate the significant results of Tukey's multiple comparison test at $\alpha=.05$. The mean values having no superscripts indicate no significant mean differences between age groups at $\alpha=.05$.

Table 8. Comparison of ear length, ear breadth, and cavum concha width by gender and ethnicity.

Ear dimension	Study	Population	Male (m)										Female (f)										Mean Difference ($M_m - M_f$)	
			n	Mean (SD)	1st	5th	10th	Percentile					n	Mean (SD)	1st	5th	10th	Percentile						
								25th	75th	90th	95th	99th						25th	75th	90th	95th	99th		
Ear length	The present study	Korean	115	64.8 (4.2)	55.4	58.2	59.6	62.6	67.2	70.9	71.8	73.8	115	61.8 (4.0)	53.1	54.7	56.4	59.4	64.9	66.9	68.0	69.3	3.0**	
	The present study	Caucasian	50	61.6 (3.8)	53.8	56.9	57.6	58.7	64.1	67.0	67.2	68.2	46	59.0 (4.3)	50.1	52.7	53.6	56.4	60.6	63.5	65.0	70.7	2.6**	
	Yoon and Jung (2002)	Korean	318	64.0 (5.0)	52.4	55.0	57.6	60.6	67.4	70.4	72.0	75.6	260	59.0 (5.0)	47.4	51.0	52.6	55.6	62.4	65.4	68.0	70.6	5.0**	
	Jung and Jung (2003)	Korean	50	65.5 (4.9)	54.1	58.2	60.0	62.0	68.3	72.8	75.5	76.9	50	60.1 (4.2)	50.3	53.6	55.0	57.0	63.3	66.0	67.5	69.9	5.4**	
	Liu (2008)	Taiwanese	100	58.4 (7.1)	41.9	45.0	49.3	55.0	62.3	67.5	69.8	74.9	100	53.9 (6.2)	39.5	41.7	46.0	50.0	57.5	61.8	61.6	68.3	4.5**	
	Bozkir et al. (2006)	Turkish	191	62.9 (3.5)	54.8	57.1	58.4	60.5	65.3	67.4	68.7	71.0	150	59.5 (3.1)	52.3	54.4	55.5	57.4	61.6	63.5	64.6	66.7	3.4**	
	Alexander and Laubach (1968)	Caucasian	500	67.1 (4.5)	57.9	60.1	61.3	63.9	70.1	72.9	74.8	77.7	-	-	-	-	-	-	-	-	-	-	-	
	Meijerman, van der Lugt, and Maat (2007)	American Dutch	919	71.0 (5.5)	58.2	62.0	63.9	67.3	74.7	78.1	80.0	83.8	434	64.0 (5.4)	-	55.1	57.1	60.4	67.6	70.9	72.9	76.6	7.0**	
	Ahmed and Omer (2015)	Sudanese	100	61.6 (3.8)	52.7	55.3	56.7	59.0	64.2	66.5	67.9	70.5	100	58.0 (3.9)	-	51.6	53.0	55.3	60.6	62.9	64.3	67.0	3.6**	
	Taura, Adamu, and Modibbo (2013)	Nigerian	137	60.8 (3.8)	51.9	54.5	55.9	58.2	63.4	65.7	67.1	69.8	82	59.5 (4.0)	-	52.9	54.4	56.8	62.1	64.5	66.0	68.7	1.3*	
Ear breadth	Porter and Olson (2001)	African American	-	-	-	-	-	-	-	-	-	-	108	57.4 (3.9)	-	51.0	52.4	54.8	60.0	62.4	63.8	66.5	-	
	Pooled value			66.7 (5.0)										60.3 (4.7)										
	Inter-population variability			12.6										11.5										
	The present study	Korean	115	33.2 (2.4)	28.7	29.4	29.7	31.5	34.9	35.9	37.0	39.8	115	29.7 (2.4)	24.4	25.7	27.0	28.2	31.2	32.7	33.8	34.9	3.5**	
	The present study	Caucasian	50	34.2 (2.5)	29.4	30.1	31.3	32.5	36.1	37.7	38.4	39.3	46	31.4 (2.6)	25.8	27.9	28.7	29.9	32.6	34.9	36.4	37.6	2.8**	
	Yoon and Jung (2002)	Korean	318	33.0 (4.0)	23.7	26.0	27.9	30.3	35.7	38.1	40.0	42.3	260	30.0 (4.0)	20.7	23.0	24.9	27.3	32.7	35.1	37.0	39.3	3.0**	
	Bozkir et al. (2006)	Turkish	191	33.1 (2.1)	28.2	29.6	30.4	31.7	34.5	35.8	36.6	38.0	150	31.2 (2.2)	26.1	27.6	28.4	29.7	32.7	34.0	34.8	36.3	1.8**	
	Alexander and Laubach (1968)	Caucasian	500	34.5 (2.9)	27.9	29.7	30.8	32.3	36.4	38.2	39.4	41.6	-	-	-	-	-	-	-	-	-	-	-	
	Meijerman, van der Lugt, and Maat (2007)	American Dutch	919	35.0 (3.3)	27.3	29.6	30.8	32.8	37.2	39.2	40.4	42.7	434	33.0 (2.9)	26.3	28.2	29.3	31.0	35.0	36.7	37.8	39.7	2.0**	
	Ahmed and Omer (2015)	Sudanese	100	34.2 (2.8)	27.8	29.7	30.7	32.4	36.1	37.7	38.7	40.6	100	32.2 (2.3)	26.9	28.4	29.3	30.7	33.8	35.2	36.0	37.6	2.0**	
Cavum concha width	Taura, Adamu, and Modibbo (2013)	Nigerian	137	30.9 (2.5)	25.0	26.7	27.6	29.2	32.5	34.1	35.0	36.7	82	29.8 (2.5)	24.1	25.8	26.7	28.2	31.5	33.0	33.9	35.5	1.1**	
	Pooled value			34.1 (3.1)										31.5 (3.0)										
	Inter-population variability			4.1										3.3										
	The present study	Korean	115	17.4 (1.9)	12.9	14.1	14.7	16.4	18.9	19.5	20.4	20.8	115	16.3 (1.8)	12.3	13.5	14.3	15.2	17.4	18.6	19.4	20.6	1.1**	
	The present study	Caucasian	50	16.8 (1.9)	12.5	13.9	14.0	15.7	18.4	18.9	19.9	20.5	46	16.0 (1.5)	12.5	13.4	14.4	15.0	17.0	17.6	18.0	19.4	0.8*	
	Bozkir et al. (2006)	Turkish	191	17.2 (1.8)	13.0	14.2	14.9	16.0	18.4	19.5	20.2	21.4	150	16.5 (1.8)	12.3	13.5	14.2	15.3	17.7	18.8	19.5	20.7	0.7**	
	Ahmed and Omer (2015)	Sudanese	100	18.7 (2.1)	13.8	15.2	16.0	17.3	20.1	21.4	22.1	23.6	100	17.6 (1.7)	13.7	14.8	15.4	16.5	18.7	19.8	20.4	21.5	1.1**	
	Pooled value			17.5 (1.9)										16.7 (1.7)										
	Inter-population variability			1.9										1.6										

* $p < 0.05$, ** $p < 0.01$.

4. Discussion

Three-dimensional ear scan images collected in this study by applying 3D scanning and ear casting techniques can be applied to designs of ear products. Images of complex shapes of concha and ear canal could be captured by scanning their cast. 3D ear images of Koreans and Caucasians collected in this study would be effectively applied to ergonomic design of various ear products including earphones, earsets, hearing aid devices, headsets having ear-band, smart glasses, and protective goggles. For example, Lee et al. (2016a) reported the use of 3D ear scan images to find the most accurate size and shape of an earphone by virtually fitting an earphone CAD design to concha images of the ears having various shapes.

The present study selected 25 ear dimensions for design of ear products by a comprehensive literature review and a recommendation by a panel of experts. Nine dimensions related to ear product design were systematically selected from the literature review. An addition of 14 new ear dimensions, which have not been measured in the previous studies, were proposed by a panel of experts by identifying the design variables of ear products which require application of ear measurements for proper fit. Then, 13 new ear landmarks were defined in this study for the selected ear dimensions. The ear dimensions and corresponding ear landmarks proposed in the present study can be effectively applied to ear anthropometric analysis and ear product design.

The centre-of-concha, a virtual landmark, was recommended in this study for ear product design by a panel of experts. Virtual landmarks have been used in previous 3D anthropometric studies (Choi, Nam, and Kim 2014; ISO/DIS 2015; Xi and Shu 2009) to align 3D images and measure anthropometric dimensions. The centre-of-concha proposed in the present study is important for its close relationship to the reference point in earphone design, which is located at the middle-central part of a earphone head to house a speaker and fit the concha. The centre-of-concha can also be applied to determine the size and shape of earphone components such as earphone-head, hook, and ear-band.

As inter-gender and inter-population variations of ear measurements were found smaller than the corresponding intra-population variations, an ear product needs to be designed mainly by considering personal differences, not inter-gender or inter-population differences. Both the inter-population variabilities (2.0–12.6 mm) and inter-gender variabilities (0.1–7.0 mm) of ear length, ear breadth, and cavum

concha width were less than the corresponding intra-population variabilities (7.5–22.2 mm). Therefore, the size category system of earphone can be determined based on a composite dataset of ear measurements of gender and populations. For example, the diameter of the earphone head can be determined as 14.6 mm for small size (coverage range = 13.4–15.7 mm), 16.9 mm for medium size (coverage range = 15.7–18.1 mm), and 19.2 mm for large size (coverage range = 18.1–20.4 mm) based on the variability of concha width of both Koreans and Caucasians ($p_{0.05} = 13.4$ mm and $p_{0.95} = 20.4$ mm) to cover 90% of the Korean and Caucasian populations.

The present study found that the height of pinna increased but the width of the concha decreased with age. Many studies, including Niemitz, Nibbrig, and Zacher (2007) and Sharma (2016), report positive growth gradients by age for ear-height dimensions such as greatest ear length, ear base length, and concha height. While Niemitz, Nibbrig, and Zacher (2007) identified approximately zero gradients for ear-width dimensions such as concha width and incisura intertragica width by analysing 1,440 ears photographed from 724 pairs of ears of Germans aged 0–92 years, the present study observed a negative growth gradient for concha width by analysing 230 Korean and 96 Caucasian right ears in their 20s–50s. Although more research is needed on the growth gradients of ear dimensions, the ear measurement studies commonly find that the ear grows mainly vertically, but not horizontally, throughout the lifetime.

Research on in-depth ear anthropometric measurement and analysis and application of 3D ear data to ear product design can be conducted in future. First, an investigation of intra- and inter-measurer variabilities can show the reliability of the ear measurement method using 3D ear scan images. Second, similarity and symmetry of the left and right ears can be quantified for an individual using 3D ear scan data. Lastly, 3D ear scan data can be applied to the development of an ergonomic design of an ear product and the virtual fit evaluation of an ear product. For example, the 3D size and shape data of concha and ear canal can be used to develop the ergonomic design of an ear product such as hearing aid and in-ear type earphone for better comfort and fit. Furthermore, 3D ear scans can be utilised to predict the comfort and fit of an ear product design to the ear by virtually fitting the ear product design to 3D ear scans in various sizes and shapes and analysing the overlapping areas between the design and the ears (Lee et al. 2016b).

Disclosure statement

No potential conflict of interest was reported by the authors.

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ORCID

Wonsup Lee  <http://orcid.org/0000-0002-8094-1251>
 Xiaopeng Yang  <http://orcid.org/0000-0002-5180-109>
 Hayoung Jung  <http://orcid.org/0000-0003-1315-2313>
 Chulwoo Kim  <http://orcid.org/0000-0001-8299-0578>
 Heecheon You  <http://orcid.org/0000-0002-1914-5774>

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